

Retained Evidence for Changing Predictions

Exact Moment Ambiguity and Fresh-Audit Learning Diagnostics

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Abstract

We investigate two distinct ways to use historical labels during structural learning: fitting future candidates and certifying their current risks. For a known finite input law, we derive the exact worst-case ambiguity of a Brier-risk change given approximate label moments. With exact moments, prediction movement outside the retained span determines matching upper and lower fresh-testing orders for symmetric prediction pairs in a stated small-margin regime. The mathematics specializes optimal recovery, control variates and sequential testing; it is not an established new general inference principle. Finite-state experiments show both the value of oracle retained evidence and the loss caused by its finite estimation error. A stress test exposes invalid adaptive reuse. Separately, 336 trained neural trajectories compare permanent audit withholding with post-decision release, informative and random growth, and same-arrival fixed-capacity online controls. Release lowers mean final risk for both proposal mechanisms. Residual proposals show no resolved advantage over random proposals, while larger fixed learners obtain lower risks on two nonlinear tasks. These results diagnose a preventable data-use restriction without proving that structural adaptation is generally superior. Complete proofs, frozen protocols, raw outcomes and explicit resource limits accompany the report.

Research status and relation to the preprint candidates

This September 2026 report contains the analysis and experiments from that research stage. Repository preparation did not rerun the experiments. The existing-results preprint candidate, *Evidence Interfaces in Repeated Neural Growth*, is prepared separately. The present report is material for scientific study within that line, with its stated assumptions and limits. The strongest new mathematical result in the coordinated investigation is a separate matching source-mass order for four-state irreversible finite representations; it belongs to the finite-representation manuscript.

The motivating distinction is practical. Releasing an old audit into future fitting does not give a new candidate independent historical certification data. Conversely, retained information need not become useless merely because a predictor changes. We quantify the latter point within a fixed moment interface and test the former in actual trained networks.

1 Model, information access, and prior work

Let X have a known law μ on $\{1, \dots, m\}$, with $\mu_i > 0$. Define

$$\langle a, b \rangle_\mu = \sum_i \mu_i a_i b_i, \quad \|a\|_p = \left(\sum_i \mu_i |a_i|^p \right)^{1/p}.$$

Labels $Y \in \{0, 1\}$ have unrestricted conditional signed mean $g_i = \mathbb{E}[2Y - 1 \mid X = i] \in [-1, 1]$. A dictionary $B \in \mathbb{R}^{m \times d}$ is fixed in advance, with columns b_j and span V . The exact population-moment interface reveals

$$m_j(g) = \langle b_j, g \rangle_\mu, \quad j = 1, \dots, d.$$

This specifies information access, not physical storage. A real scalar can encode a finite label vector, and even a fixed empirical linear sketch with rationally independent weights can encode integer count vectors. No memory lower bound follows from the dictionary rank. Moreover, the distribution of an empirical summary can contain information absent from its population expectation; our exact-moment lower bound does not automatically apply to that distribution.

For predictions $p, q \in [0, 1]^m$, write $h = q - p$ and $c = \langle h, 1 - p - q \rangle_\mu$. Expanding squares yields

$$\Delta_g(p, q) := \mathbb{E}[(Y - p(X))^2 - (Y - q(X))^2] = c + \langle h, g \rangle_\mu. \quad (1)$$

Only the change h multiplies the unknown label means. The known input law makes c and the approximation geometry computable. Unknown μ requires additional uncertainty control.

Control-variate confidence intervals already estimate the residual after subtracting a variable with known mean [1]; projection-based residual concentration and coefficient-error analysis are established [2, Eq. (6), Theorem 3.1, Appendix B]. Uniform confidence regions accommodate adaptive use [3, Theorem 3, Corollary 1]. In structural learning, Amoukou et al. already certify current prediction-risk differences by minimizing over a stationary probability region [4, Lemma A.13]. Our analysis specializes these tools and a standard testing lower bound [5, Lemma 1] to retained label moments. Convex duality is used in its standard form [6].

2 Sharp reconstruction from approximate moments

Suppose $z \in \mathbb{R}^d$ is known to satisfy $|z_j - m_j(g)| \leq e_j$, where $e_j \geq 0$. For fixed h , let

$$\mathcal{R}(h, B, e) = \inf_T \sup_{\substack{g \in [-1, 1]^m \\ |z - m(g)| \leq e}} |T(z) - \langle h, g \rangle_\mu|,$$

where the supremum includes all admissible (g, z) and the infimum allows arbitrary functions T , including nonlinear ones.

Theorem 2.1 (Exact ambiguity of retained moments). *One has*

$$\mathcal{R}(h, B, e) = \max_{\substack{\|u\|_\infty \leq 1 \\ |B^\top \text{diag}(\mu)u| \leq e}} \langle h, u \rangle_\mu \quad (2)$$

$$= \min_{a \in \mathbb{R}^d} \left\{ \|h - Ba\|_1 + \sum_j e_j |a_j| \right\}. \quad (3)$$

In particular, $\mathcal{R}(h, B, 0) = \text{dist}_{L_1(\mu)}(h, V)$, which is zero if and only if $h \in V$.

Proof. For any a , the estimator $T(z) = a^\top z$ obeys

$$|T(z) - \langle h, g \rangle_\mu| \leq \|h - Ba\|_1 + \sum_j e_j |a_j|.$$

For a lower bound take $z = 0$. Every feasible u in (2), as well as $-u$, is consistent with $z = 0$. Their target values are opposite, so one of the two errors of any $T(0)$ is at least $|\langle h, u \rangle_\mu|$. The feasible set is nonempty, symmetric and compact.

It remains to equate (2) and (3). Assign nonnegative multipliers s, t to the upper and lower moment constraints in the maximization. Maximizing its Lagrangian over $[-1, 1]^m$ yields

$$\|h - B(s - t)\|_1 + e^\top (s + t).$$

For fixed $a = s - t$, the smallest second term is $\sum_j e_j |a_j|$, attained by the positive and negative parts of a . Finite linear-program strong duality applies: the primal is feasible with finite optimum and the dual is feasible, for example at $a = 0$. This proves equality and attainment. Finally, finite-dimensional V is closed, so its L_1 distance is zero exactly on V . Adding the known c in (1) does not change any error. $\square$

Corollary 2.2 (No-data decision obstruction). *Let $\|h\|_\infty \leq 1$ and $h \notin V$. Symmetric predictions $p = (1 - h)/2$, $q = (1 + h)/2$ admit two binary-label laws with identical exact retained moments and strictly opposite gains. No randomized rule based solely on these moments can have type-I error at most α and power at least $1 - \beta$ on both laws when $\alpha + \beta < 1$.*

Proof. With $e = 0$, choose a maximizer u of (2). Set $g = \pm u$. Both give zero moments, while $c = 0$ and the gains are $\pm \text{dist}_{L_1(\mu)}(h, V)$. The acceptance probability is the same under both indistinguishable observations; the two requirements would give $1 - \beta \leq \alpha$. $\square$

3 The additional cost of fresh evidence

Let v be the $L_2(\mu)$ projection of fixed h onto V , and put

$$r = h - v, \quad S = \|r\|_2^2, \quad M = \|r\|_\infty.$$

Exact moments determine $A = c + \langle v, g \rangle_\mu$. On n fresh iid full pairs, define

$$\hat{\Delta} = A + \frac{1}{n} \sum_{i=1}^n r(X_i)(2Y_i - 1).$$

Theorem 3.1 (Residual upper bound). *For $u > 0$ and $n \geq 1$, let*

$$F_n(u) = \sqrt{\frac{2Su}{n}} + \frac{4Mu}{3n}.$$

Then $\mathbb{P}(|\hat{\Delta} - \Delta| > F_n(u)) \leq 2e^{-u}$, with failure probability e^{-u} for either one-sided endpoint. Accepting when $\hat{\Delta} > F_n(u)$ has type-I error at most e^{-u} for $\Delta \leq 0$. For $\Delta \geq \gamma > 0$, it has power at least $1 - e^{-u}$ whenever

$$n > \max \left\{ \frac{32Su}{\gamma^2}, \frac{16Mu}{3\gamma} \right\}. \quad (4)$$

If $S = 0$, the gain is exactly known and no fresh data are required.

Proof. If $S = 0$, positivity of every μ_i implies $r = 0$, so the gain is exactly known. Assume $S > 0$, hence $M > 0$. Write $W = r(X)(2Y - 1)$ and $Z = W - \mathbb{E}W$. Then $\mathbb{E}W^2 = S$, $\text{Var}(W) \leq S$, and $|Z| \leq 2M$. For $k \geq 2$, $\mathbb{E}|Z|^k \leq (2M)^{k-2} \mathbb{E}Z^2$. The exponential series, $k! \geq 2 \cdot 3^{k-2}$, and $\log(1 + x) \leq x$ give

$$\log \mathbb{E}e^{\lambda Z} \leq \frac{\lambda^2 S}{2(1 - 2M|\lambda|/3)}, \quad |\lambda| < 3/(2M).$$

For $t > 0$ choose $\lambda = t/(S + 2Mt/3)$ in the exponential Markov bound for the sum of n independent copies, obtaining

$$\mathbb{P} \left(\frac{1}{n} \sum_i Z_i \geq t \right) \leq \exp \left\{ -\frac{nt^2}{2(S + 2Mt/3)} \right\}.$$

Substitution of $F_n(u)$ makes the exponent at least u . Repeat for $-Z$. The strict sample condition (4) makes both terms of F_n less than $\gamma/4$. On the lower-tail coverage event, $\hat{\Delta} \geq \Delta - F_n > F_n$. If $S = 0$, positivity of every μ_i implies $r = 0$, proving the final statement. $\square$

Theorem 3.2 (Matching residual lower bound). *Suppose $\|h\|_\infty \leq 1$, $S > 0$, and*

$$0 < \gamma \leq \frac{S}{2M}, \quad \alpha, \beta \in (0, 1), \quad \alpha + \beta < 1.$$

Use symmetric predictions $p = (1 - h)/2$, $q = (1 + h)/2$. A test receives exact V -moments, then observes a fresh iid stream of full (X, Y) pairs, and may randomize and stop sequentially. Suppose it stops almost surely under the laws constructed below, has acceptance probability at most α under every law with $\Delta \leq 0$, and at least $1 - \beta$ under every law with $\Delta \geq \gamma$. There is a law with gain exactly γ under which

$$\mathbb{E}N \geq \text{kl}(1 - \beta, \alpha) \frac{S}{\gamma^2}. \quad (5)$$

Proof. Under P_0 set $g = 0$; under P_1 set $g = \gamma r/S$. The size condition gives $|g_i| \leq 1/2$, so both laws have conditional Bernoulli means in $[1/4, 3/4]$. Orthogonality gives $\langle b_j, r \rangle_\mu = 0$, so the exact initial moment vector is zero under both laws. Also $c = 0$ and

$$\Delta_0 = 0, \quad \Delta_1 = \frac{\gamma}{S} \langle h, r \rangle_\mu = \gamma,$$

using $\langle h, r \rangle_\mu = \|r\|_2^2 = S$.

The elementary inequality $\text{kl}(a, b) \leq (a - b)^2/[b(1 - b)]$ follows from applying $\log x \leq x - 1$ to the two logarithms. Therefore

$$\begin{aligned} \text{KL}(P_1, P_0) &= \sum_i \mu_i \text{kl}\left(\frac{1 + \gamma r_i/S}{2}, \frac{1}{2}\right) \\ &\leq \sum_i \mu_i \left(\frac{\gamma r_i}{S}\right)^2 = \frac{\gamma^2}{S}. \end{aligned}$$

If $\mathbb{E}_1 N = \infty$, the claim is immediate. Otherwise, the stopped log-likelihood ratio equals

$$\sum_{i \geq 1} \mathbf{1}_{\{N \geq i\}} \log \frac{dP_1}{dP_0}(X_i, Y_i).$$

The likelihood increments are bounded, the expectation of N is finite, and $\{N \geq i\}$ is measurable before draw i . Absolute integrability and conditional expectation show that the stopped-transcript divergence equals $\mathbb{E}_1 N \text{KL}(P_1, P_0)$. The common stopping and randomization rule contributes no additional likelihood factor. By data processing onto the acceptance event and binary-KL monotonicity,

$$\mathbb{E}_1 N \text{KL}(P_1, P_0) \geq \text{kl}(P_1(\text{accept}), P_0(\text{accept})) \geq \text{kl}(1 - \beta, \alpha).$$

Combining the inequalities proves (5). □

For $\alpha = \beta = \delta \leq 1/4$, the binary divergence is $(1 - 2\delta) \log((1 - \delta)/\delta)$, of order $\log(1/\delta)$. The small-margin condition gives $M/\gamma \leq S/(2\gamma^2)$, absorbing the range term in the upper bound. Thus, for symmetric prediction pairs in this small-margin regime, the minimax additional testing cost for this fixed exact-moment interface has order

$$S\gamma^{-2} \log(1/\delta).$$

The lower construction is a problem-specific specialization of the standard sequential change-of-measure method, not a new testing principle. It does not allow arbitrary additional informative training histories: their divergence could pay for part of the decision.

4 Finite historical evidence and adaptive pairs

Fix the dictionary before an old iid sample H of size N_0 , and assume $|b_j| \leq B_j$. Retain

$$z_j = \frac{1}{N_0} \sum_{i=1}^{N_0} b_j(X_i)(2Y_i - 1), \quad e_j = B_j \sqrt{\frac{2 \log(2d/\delta_0)}{N_0}}.$$

Hoeffding's inequality and a union bound give probability at least $1 - \delta_0$ for the event $\mathcal{E}_0$ on which every $|z_j - m_j(g)| \leq e_j$. All future fitting and pair selection may depend on H .

At event t , condition on pre-audit history and select p_t, q_t, a_t, n_t . Put $v_t = Ba_t$, $r_t = q_t - p_t - v_t$, $c_t = \langle q_t - p_t, 1 - p_t - q_t \rangle_\mu$, and

$$H_t = \sum_j e_j |a_{tj}|, \quad F_t = \sqrt{\frac{2 \|r_t\|_2^2 \log(1/\delta_t)}{n_t}} + \frac{4 \|r_t\|_\infty \log(1/\delta_t)}{3n_t}.$$

Using a new independent audit, define

$$L_t = c_t + a_t^\top z + \frac{1}{n_t} \sum_{i=1}^{n_t} r_t(X_{ti})(2Y_{ti} - 1) - H_t - F_t.$$

Proposition 4.1 (Adaptive transfer with historical uncertainty). *For a finite or countable sequence of such events,*

$$\mathbb{P}(\exists t : L_t > \Delta_t) \leq \delta_0 + \sum_t \delta_t.$$

Proof. On $\mathcal{E}_0$, the deterministic inequality $|a^\top(z - m(g))| \leq \sum_j e_j |a_j|$ holds for every a , hence for adaptive a_t . Conditional on each pre-audit history, the residual is fixed and Theorem 3.1 bounds the fresh one-sided failure by δ_t . Integrate and take a union bound together with $\mathcal{E}_0$. Countable unions follow by increasing finite unions. $\square$

The proposition is a standard simultaneous-region argument, with a direct structural-learning precedent [4, Lemma A.13]. It transfers estimates together with their uncertainty, not old betting products. New dictionary columns fitted on old labels require a uniform class argument or fresh data. Completed audits may later enter fitting while future audit blocks remain independent. Refreshing the inference region across update times needs simultaneous bounds for those updates. When historical errors are nonzero, the L_2 projection need not minimize $H_t + F_t$; shrinking a_t , including to zero, can be preferable. If $r_t = 0$, residual-only future auditing cannot shrink H_t .

5 Releasing completed audits into later fitting

Proposition 5.1 (Standard conditional-validity corollary). *Let $\mathcal{H}_{t-1}$ contain every earlier fitting sample, completed audit, decision and optimizer state. Before revealing audit A_t , construct the incumbent and all candidate predictors measurably from $\mathcal{H}_{t-1}$ and independent proposal randomness. Suppose A_t is iid from the task law, independent of the complete pre-audit sigma field, including proposal randomness, and the conditional probability of a falsely accepted comparison is at most α_{tj} , with predictable total $\sum_{t,j} \alpha_{tj} \leq \delta$. Commit only the chosen frozen candidate, or hold the incumbent exactly. Then with probability at least $1 - \delta$ every accepted inequality holds in population. After committing, A_t may enter $\mathcal{H}_t$ and any future fitting pool.*

Proof. Condition on the full pre-audit history and proposal randomness. Current predictors are fixed and the audit remains independent, so the stipulated conditional error bound applies. Integrate over history and sum the error bounds. This yields a simultaneous event on which every installed change satisfies its audited inequality. Label release changes later history, not the conditional argument for a genuinely fresh later block. A hold changes neither predictor nor risk. Increasing finite unions cover a countable sequence. No candidate-discovery or power statement follows. $\square$

This is the same standard fresh-evidence reasoning used by the inherited controller; the new item is its implemented diagnostic. A silent refit of the installed predictor would break the claimed checkpoint protection. Optimizer state may persist, but its resulting fitted parameters must still be audited before installation. Ordinary ungated online learners and diagnostic training tails pursue a different objective and have no such guarantee.

Why a previous betting product is not a new-pair certificate. Consider a single input and $Y \sim \text{Bernoulli}(3/4)$. The old pair $p = 1/4, q = 3/4$ has difference $D = \frac{1}{2}(2Y - 1)$ and positive gain. Its product with bet $1/2$ has positive expected log increment $\frac{3}{4} \log(5/4) + \frac{1}{4} \log(3/4) > 0$ and therefore crosses any fixed threshold almost surely. Swapping the pair makes the true gain negative. Using the old product as its initial evidence would accept the harmful new pair with probability one if no new evidence were required. Even after one fresh factor, choosing the old threshold larger by $4/3$ preserves that counterexample. This does not contradict a prequential test whose null bounds every conditional increment; that is a different hypothesis. The valid transfers above retain a confidence region with its uncertainty.

6 Finite-state experiments on retained evidence

6.1 Frozen design and supplemental controls

The R1 protocol fixes a uniform four-state input $(f_1, f_2) \in \{-1, 1\}^2$, $h = 0.2(\sqrt{1-z}f_1 + \sqrt{z}f_2)$, and symmetric predictions $p = (1-h)/2, q = (1+h)/2$. The retained span is $V = \text{span}(f_1)$. The squared total movement is 0.04 and residual movement $0.04z$ for $z \in \{1, 1/4, 1/16, 1/64\}$. The null has $\eta = 1/2$; alternatives have $\eta = 1/2 + 0.003f_2/(0.4\sqrt{z})$, giving the same gain 0.003. All probabilities lie in $[1/4, 3/4]$.

For each condition, 1,000 independent replicates generate nested iid count samples at six fixed fresh sizes from 128 to 131,072. Methods use a direct paired Bernstein bound, a residual bound with an exact oracle moment, or a residual bound with 32,768 independent historical labels and a Hoeffding moment interval. The finite-history method divides error 0.05 equally between historical and fresh uncertainty. The oracle’s exact moment is privileged information, not a free finite dataset. Displayed rejection rates are pointwise at fixed sizes; the plotted grid is not a valid repeated-look test without additional allocation.

An explicitly post-hoc supplement adds two necessary controls from the unchanged archived counts. Because the R1 pair is precommitted, a direct comparison can use all old and fresh labels. A second control selects a shrunken retained coefficient by minimizing the analytic historical-plus-fresh radius using only geometry and sample sizes, before inspecting observed values. These controls preserve the frozen results and do not create a new confirmatory comparison.

At $z = 1/64$ and 8,192 fresh labels, direct testing accepts in 129 of 1,000 alternative replicates, exact-moment residual testing in 1,000, and the finite-history method in 168. The oracle result verifies the predicted mechanism, while the finite-history result prevents claiming the same efficiency for a realizable procedure. The full-pool comparator also makes clear that compressing history can lose useful evidence. A fixed imperfect retained projection leaves historical uncertainty even when residual evidence becomes abundant; shrinking toward zero avoids that particular floor.

6.2 Adaptive reuse stress test

R2 has $m \in \{4, 16, 64\}$ equiprobable inputs and the null law $\eta = 1/2$. After 4,096 historical observations, it chooses $h_j = 0.4 \text{sign}(\hat{\mathbb{E}}[1_{X=j}(Y - 1/2)])$ and symmetric predictions. Every selected pair has true gain zero. Across 1,000 new replicates, we compare naive fixed-pair Bernstein reuse, a simultaneous coordinate moment region, and 4,096 genuinely fresh labels

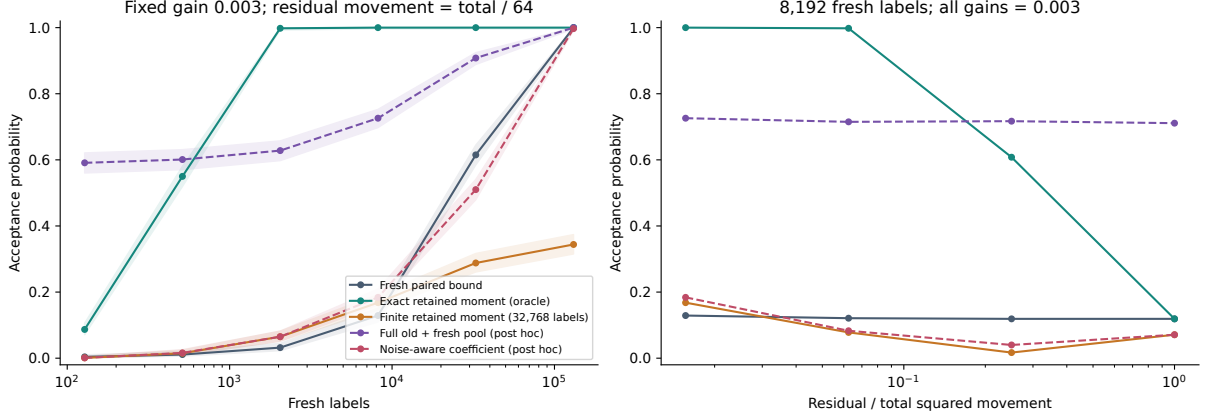


Figure 1: R1: genuine finite-state sampling, not neural training. Bands are pointwise 95% Wilson intervals over 1,000 independent replicates. Dashed methods were added after viewing the frozen results. Full-pool and finite history methods both use 32,768 additional historical labels; the oracle moment is a separate ideal-information resource.

conditional on the selected pair. The moment region is deliberately elementary and can be conservative; this null stress test makes no power claim for it.

Naive reuse falsely accepts 69, 880 and 1,000 times as m increases; fresh audits accept 3, 6 and 2 times, and the simultaneous region accepts zero. Zero observed failures do not prove zero error probability. The experiment is a designed selection-bias diagnostic, not a claim that typical neural training exhibits those false-acceptance rates.

7 Completed audits as future fitting data

The following new-seed diagnostic asks whether permanently excluding completed audits from later fitting creates avoidable limitations in a trained adaptive learner. It does not claim a new testing theorem, and it does not implement transfer of historical certification evidence to a changed prediction pair. The broad motivation is useful structural adaptation under explicit data and computation costs; success of the residual proposal is not assumed.

7.1 Implemented evidence mechanism

Let H_{t-1} contain every earlier fitting/audit observation, decision, parameter and optimizer state. Current proposal randomness may be included before the next audit. Suppose all current predictors and comparison margins are then fixed and the current block A_t is independent and iid. For old minus new bounded-loss differences $D_i \in [-1, 1]$, define

$$E_n = \frac{1}{5} \sum_{\lambda \in \Lambda_\tau} \prod_{i=1}^n \{1 + \lambda(D_i - \tau)\}, \quad \Lambda_\tau = \{.05, .1, .2, .5, (1 + \tau)^{-1}\},$$

with the implemented margins $\tau \in \{0, .0005\}$. Accept the inequality when $E_n \geq 1/\alpha_{t,j}$. This is an established bounded-mean betting construction, not a new test; the direct foundation is Waudby-Smith and Ramdas, Section 4 and Appendix A.3 of the inspected arXiv version 7 [7].

Each product has nonnegative factors and conditional expectation at most one under its null. Its fixed average therefore gives the conditional error bound needed by the preceding proposition. Every changed prediction pair starts a new product. The current experiment uses fixed blocks, with no uncorrected multiple inspections within a block.

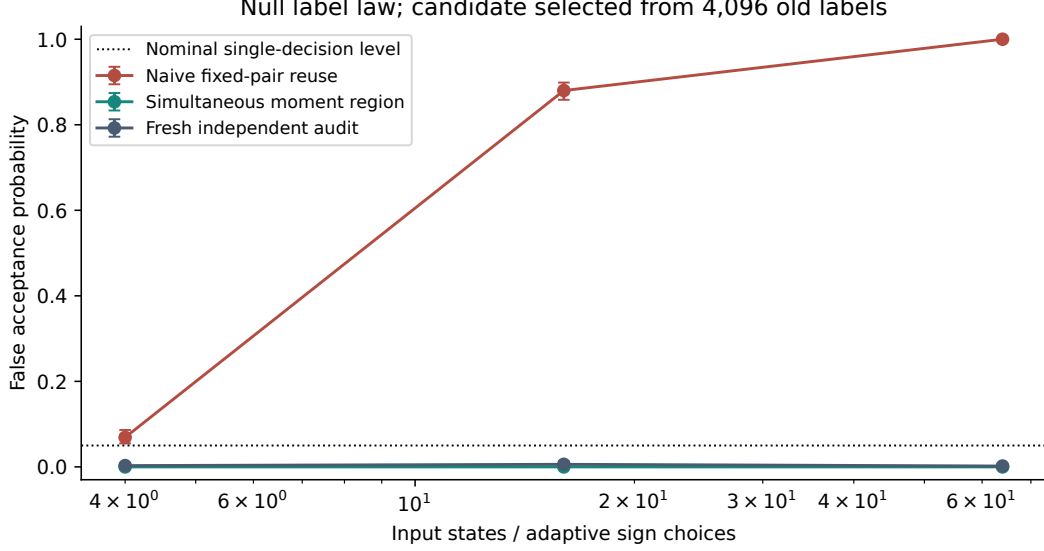


Figure 2: R2: the new candidate is chosen using the old labels. Error bars are Wilson intervals across independent replicates. Ordinary fixed-pair coverage cannot be applied retrospectively to that selected comparison.

7.2 Frozen design and actual execution

The protocol and runner hashes were frozen at 03:45:05 UTC on 15 September 2026, before seeds 750000–750011 were run. Four tasks and 12 independent seeds per task give 48 paired configurations, seven methods per configuration, 336 complete trajectories and 8,064 events. All completed the preset 24-event horizon. The main run took 330.7 seconds with two local CPU workers. An unchanged-runner recovery of the earlier interaction seed 63000 matched its full two-method history and controls before this study.

Write $\sigma(u) = 1/(1+e^{-u})$. All tasks have $X \sim N(0, I_6)$ and $Y \mid X \sim \text{Bernoulli}(.05 + .9\sigma(g(X)))$, where

$$\begin{aligned}
g_{\text{null}}(x) &= 0, \\
g_{\text{additive}}(x) &= 2.5 \tanh(2x_1 + .6) - 2 \tanh(2x_2 - .5) + 1.7 \tanh(2x_3 + .3) - 1.5 \tanh(2x_4), \\
g_{\text{interaction}}(x) &= 4 \tanh(2x_1) \tanh(2x_2) + 2 \tanh(2x_3) \tanh(2x_4), \\
g_{\text{radial}}(x) &= 3(x_1^2 + x_2^2 - 1.4).
\end{aligned}$$

The radial boundary is an additional task with four nuisance coordinates, not an instance of a supplied-module theorem. The fitted network is $p_\theta(x) = \sigma(c + \sum_{j=1}^k a_j \tanh(w_j^\top x + b_j))$ with Brier loss; all its parameters train. There are 2,048 initial fitting labels and 4,096 fresh audit labels at each event, hence 100,352 unique observed labels per method. Conditional label noise is integrated on 32,768 independent evaluator inputs per seed; the risk estimate remains Monte Carlo in X . The evaluator is called only after an entire method’s action sequence is fixed and is absent from the learner’s arguments.

Four adaptive arms cross withheld/released audit data with residual/random proposals. The residual score is $(\bar{r}\bar{z})^2/(\bar{z}^2 + 10^{-3})$, where $r = y - p$ and $z = p(1 - p) \tanh(w^\top x + b)$. Four incoming-vector candidates receive 40 minibatch ascent updates, and one is selected on at most 4,096 historical fitting points. Random additions use the same initial candidate distribution and random selection. Both add a unit with outgoing weight zero, preserving the current predictor before branch fitting. Each event fits both continuation and growth from the incumbent. Growth is accepted only after its gains over both incumbent and continuation exceed .0005; otherwise a certified positive continuation is accepted, or the incumbent is held exactly.

Task	Release: residual	Release: random	Residual minus random
Null	-1.994 [-2.813, -1.176]	-2.016 [-2.846, -1.186]	0.022 [-0.054, 0.097]
Additive	-4.576 [-6.648, -2.505]	-3.943 [-5.666, -2.221]	0.063 [-0.237, 0.363]
Interaction	-3.721 [-4.557, -2.885]	-4.657 [-6.277, -3.038]	-0.110 [-0.744, 0.524]
Radial	-1.142 [-2.131, -0.152]	-1.199 [-2.081, -0.318]	0.554 [-0.171, 1.280]

Table 1: Final Brier-risk differences, in units of 10^{-3} , with descriptive 95% paired intervals. “Release” means released minus withheld. The last column compares proposals under release. Negative values favor the first method. All four release-by-proposal interaction intervals also contain zero.

Three controls receive the same label arrivals and release times: persistent online learners at fixed widths two and 26, and a fixed width-26 learner using a fresh gate on continuation. Width 26 equals the adaptive maximum $2 + 24$ and is specified before outcomes. The ungated controls make no checkpoint-monotonicity claim. The adaptive arms allocate $.05/(3 \cdot 24)$ to each comparison and the one-comparison fixed gate $.05/24$, both spending the same trajectory error budget. That budget is not a simultaneous guarantee across experimental methods and seeds.

All arms receive six million parameter-example updates for warmup and eight million per event, including candidate search and rejected branches, with small integer remainders recorded. They use persistent Adam ($.9, .999$), learning rate $.01$, denominator constant 10^{-8} and minibatches of at most 256. Proposal ascent uses learning rate $.04$. Branches clone the incumbent’s optimizer state; only the selected branch commits it. New-unit moments start at zero and retain the existing step counter. Random proposals allocate saved search work to branch fitting. Fixed learners devote the entire event budget to continuation. The 198-million-update horizon and 32-million-update ungated tail match an operation proxy, not FLOPs, optimizer steps, memory or wall time. These shared persistent-minibatch conventions differ from the earlier full-batch/reset runner; the new factorial comparison isolates release within the new protocol.

7.3 Observed release effects and negative controls

Table 1 reports primary paired effects. Confidence intervals are descriptive 95% Student- t intervals across the 12 independent seeds; there is no multiplicity adjustment. Event time points and proposals are not treated as independent replications.

Both proposal mechanisms benefit from release in the four measured means. Their released comparison does not resolve a residual-proposal advantage; intervals containing zero do not establish equivalence. Mean released residual final risks are $.250716$, $.131908$, $.144304$ and $.121427$ on null, additive, interaction and radial, with respective widths two, four, six and three. Permanently withheld residual risks are $.252710$, $.136484$, $.148025$ and $.122568$. Improvement therefore need not be accompanied by additional growth: mean interaction width remains six under both release policies.

The fixed width-26 online learner has final mean risks $.250857$, $.131491$, $.137553$ and $.117145$. Released residual minus this control is $+.006752$ $[.006203, .007300]$ on interaction and $+.004282$ $[.003656, .004907]$ on radial. Even the gated fixed architecture outperforms released residual in those comparisons: $+.003801$ $[.003225, .004378]$ and $+.002762$ $[.002014, .003509]$, respectively. Thus removal of the gate is not the sole explanation for these fixed-capacity results. Fixed width two is strong on null (mean $.250107$) but underfits interaction ($.209650$) and radial ($.192170$). The larger control is needed for credible attribution.

After the preset ungated tail, released residual risks become $.250161$, $.131254$, $.142542$ and $.118292$. The release-minus-withheld effects remain negative, but the interaction/radial gaps to fixed26 online remain $+.004743$ $[.004021, .005464]$ and $+.001445$ $[.001061, .001829]$. On additive, this post-training comparison instead favors the smaller released residual network by $-.000329$ $[-.000491, -.000167]$. These tails are diagnostics, not protected deployment updates.

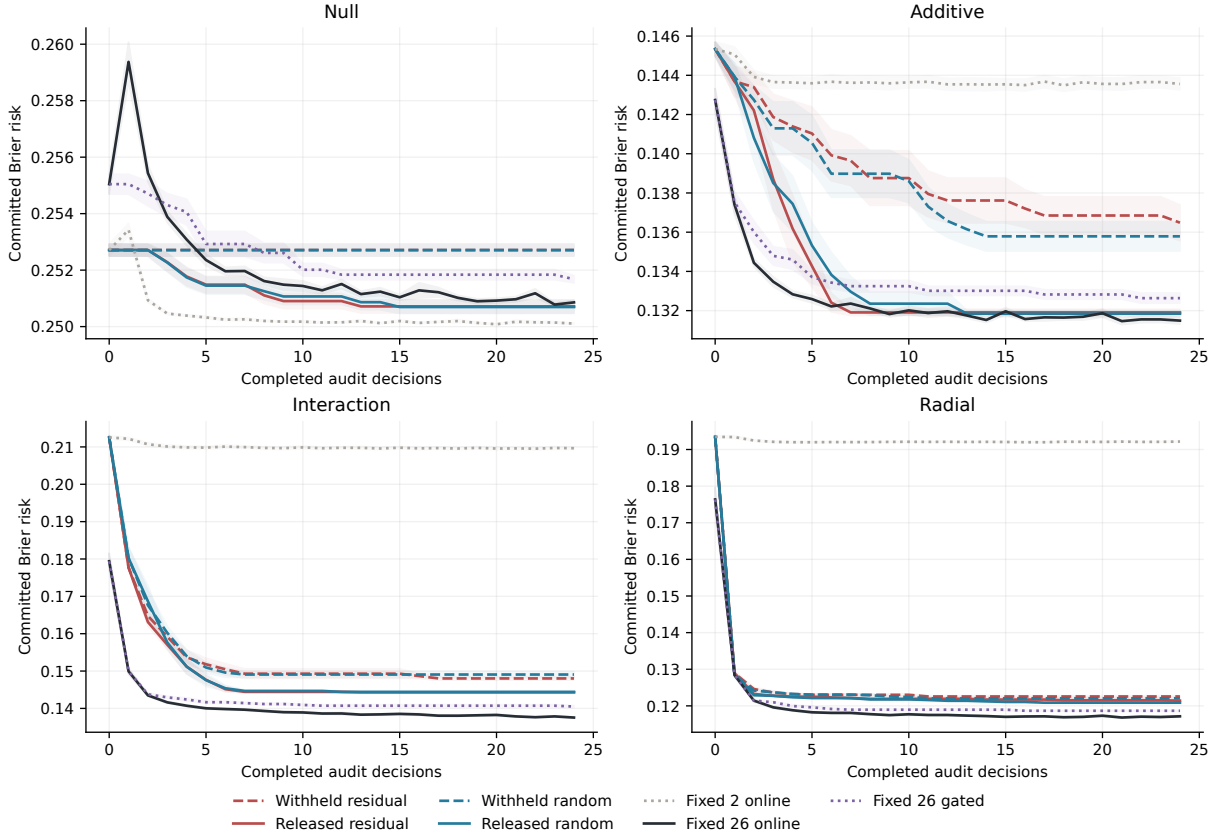


Figure 3: Actual 24-event histories; curves are seed means and bands are ± 1 standard error. Released labels improve the adaptive arms, while fixed-capacity controls expose both underfitting at width two and stronger risk performance at width 26. All arms share data arrivals and optimization proxy budgets.

All adverse and favorable task outcomes are retained.

7.4 Resource ledger and verification

Only 96,256 labels have been released before fitting the final online candidates; the last audit is released after that decision. Withheld arms fit on exactly 2,048 distinct labels. Released adaptive arms fit on roughly 96,061–96,256, while fixed26 fits about 90,625–90,644 within its matched parameter–example budget. Repeated fit accesses still differ: released residual averages 9.84m, 6.36m, 4.71m and 7.31m across tasks, versus .947m for fixed26 and 11.647m for fixed2. Pool availability is not silently equated with actual fitting access.

Residual candidate optimization costs 6.881m parameter–example updates over the horizon. Rejected branch fits alone average 168.727m–182.226m for released residual. Audit prediction work adds 5.800m–14.090m parameter–example evaluations, versus 41.091m for fixed26 gated. Adaptive gates perform 1,474,560 betting-factor evaluations per trajectory; fixed26 gated performs 491,520. Prediction and selection work is logged separately. The adaptive networks have substantially lower inference capacity than fixed26, so lower fixed-model risk is a resource tradeoff rather than a universal dominance statement.

Independent reconstruction from the raw data and saved branch parameters checks all 8,064 events, 14,976 betting comparisons and evaluator risks; maximum log-e discrepancy is 9.95×10^{-13} . All 48 run hashes, 336 access ledgers and four frozen files pass integrity checks. A separate full seven-arm interaction replay matches all 168 events at relative tolerance 10^{-10} and absolute tolerance 10^{-12} , excluding elapsed times. Checks also verify gradients, function-

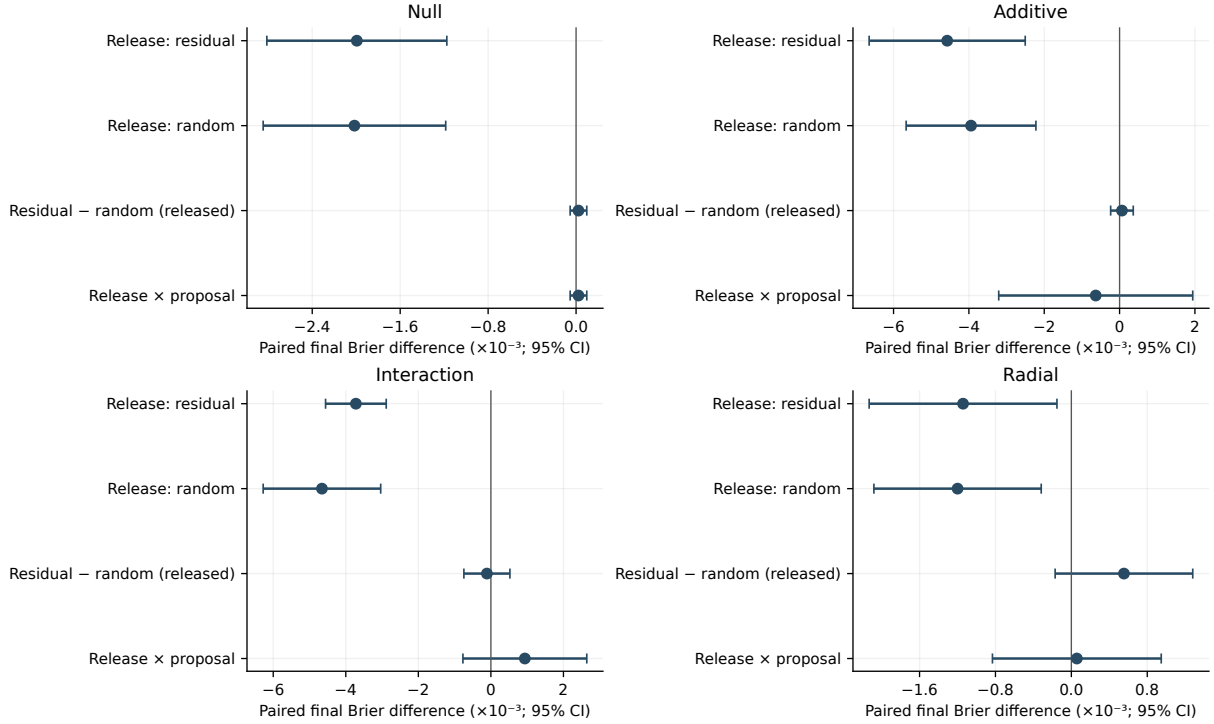


Figure 4: Prespecified paired final-risk contrasts. Bars are descriptive 95% intervals across independent seeds, without multiplicity correction. The scale is 10^{-3} Brier-risk units.

preserving additions, frozen candidate hashes, immutable commitments, post-decision release and current/future-label perturbations. No gated checkpoint increases the evaluator’s estimated risk in this sample; that observation is not the conditional proof.

Scientific status. The study diagnoses avoidable permanent withholding under a controlled learner and supplies same-arrival online controls. It does not demonstrate a new information score, retained certification evidence, optimized resource allocation or a practical deep-learning advantage. Synthetic scope, finite horizons, constant optimizer rates and proxy-compute matching remain limitations. Complete event and run summaries, a representative JSON/NPZ trajectory, commands, the frozen protocol, and plotting/audit code are in the repository. Other raw JSON/NPZ outcomes are listed by hash in the local-archive manifest. It is an experimental component of the evidence/learning investigation.

8 Verification, contribution and remaining limits

The finite-moment arguments have complete proofs. An independent numerical check compared primal and dual programs on 100 seeded instances and checked the invisible posterior, gain and KL construction. Maximum discrepancies were below 5.1×10^{-16} . Such computations check consistency and counterexample attempts, not general proof validity.

The experiments support two bounded conclusions: permanent audit withholding can impede later fitting, and historical certification information must be retained with a valid uncertainty argument. Neither result establishes an effective entropy-based controller, general neural architecture optimality, or a resource advantage over strong fixed learners. Exact population moments, known finite input distributions and precommitted dictionaries are substantial assumptions. New adaptive dictionary columns need a different uniform-inference argument. The matching residual testing result for symmetric pairs in the stated small-margin regime prices additional evidence

Task	Method	Final risk	Tail risk	Width
Null	Withheld residual	0.25271	0.25410	2.00
Null	Released residual	0.25072	0.25016	2.00
Null	Withheld random	0.25271	0.25410	2.00
Null	Released random	0.25069	0.25015	2.00
Null	Fixed 2 online	0.25011	0.25022	2.00
Null	Fixed 26 online	0.25086	0.25083	26.00
Null	Fixed 26 gated	0.25168	0.25088	26.00
Additive	Withheld residual	0.13648	0.13700	3.67
Additive	Released residual	0.13191	0.13125	4.00
Additive	Withheld random	0.13579	0.13622	3.83
Additive	Released random	0.13185	0.13125	4.00
Additive	Fixed 2 online	0.14356	0.14358	2.00
Additive	Fixed 26 online	0.13149	0.13158	26.00
Additive	Fixed 26 gated	0.13264	0.13167	26.00
Interaction	Withheld residual	0.14803	0.14711	6.00
Interaction	Released residual	0.14430	0.14254	6.00
Interaction	Withheld random	0.14907	0.14833	5.92
Interaction	Released random	0.14441	0.14244	6.00
Interaction	Fixed 2 online	0.20965	0.20959	2.00
Interaction	Fixed 26 online	0.13755	0.13780	26.00
Interaction	Fixed 26 gated	0.14050	0.13899	26.00
Radial	Withheld residual	0.12257	0.11954	3.00
Radial	Released residual	0.12143	0.11829	3.00
Radial	Withheld random	0.12207	0.11951	3.00
Radial	Released random	0.12087	0.11828	3.00
Radial	Fixed 2 online	0.19217	0.19210	2.00
Radial	Fixed 26 online	0.11714	0.11685	26.00
Radial	Fixed 26 gated	0.11867	0.11731	26.00

Table 2: Complete mean outcomes of all seven arms. Tail risk follows a separate ungated training budget and is outside the checkpoint guarantee. Values are generated directly from saved results; uncertainty and paired comparisons are reported above and in the raw CSV summaries.

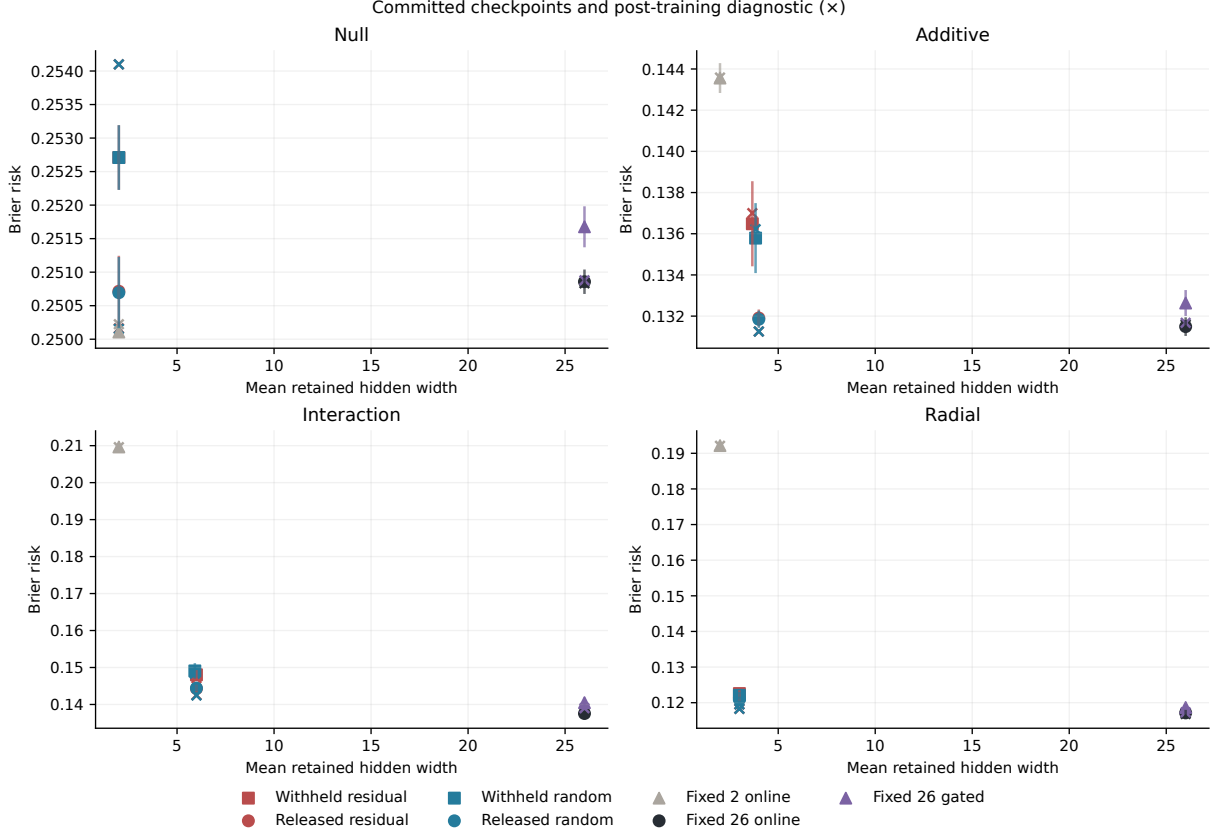


Figure 5: Risk and retained capacity. Filled markers show final committed mean risks with 95% intervals; crosses show means after the additional ungated tail. The fixed26 risk advantage on harder tasks comes with larger inference capacity.

after the specified interface, not the cost of acquiring arbitrary training history. Novelty of the underlying statistical tools is explicitly excluded.

The repository includes the scientific source, configurations, all reported summary outcomes, and a representative complete neural trajectory. Large raw files remain in the local archive; their sizes and hashes are listed in the repository raw-artifact manifest. Historical execution and replay records are distinguished from repository checks. Author verification, external peer review, and publication readiness are not asserted.

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